SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Hemalakshmi H (192571049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

---

B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

---

C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

---

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High

- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

☑ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

---

B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Q1 Answer:

A. Handling missing values

A.i Identify missing values

Sample dataset (focused columns):

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Missing values:

- **Age:** C2
- **Monthly_Spend:** C3
- **Purchase_Frequency:** C5

A.ii Median imputation (first technique)

We compute medians using available (non-missing) values.

Age (non-missing): 23, 31, 105, 27, 40, 36, 42 Sorted: 23, 27, 31, 36, 40, 42, 105 → median = 36

So:

- **Impute Age for C2 = 36**

Monthly Spend (non-missing): 20, 35, 500, 25, 45, 38, 300 Sorted: 20, 25, 35, 38, 45, 300, 500 → median = 38

So:

- **Impute Monthly_Spend for C3 = 38**

Purchase_Frequency (non-missing): 5, 8, 10, 50, 12, 9, 45 Sorted: 5, 8, 9, 10, 12, 45, 50 → median = 10

So:

- **Impute Purchase_Frequency for C5 = 10**

Imputed dataset (key columns):

Customer_ID	Age	Monthly_Spend (₹K)	Purchase_Frequency
C1	23	20	5
C2	36	35	8
C3	31	38	10
C4	105	500	50
C5	27	25	10
C6	40	45	12
C7	36	38	9
C8	42	300	45

Justification (median):

- **Median is robust to outliers** (e.g., Age = 105, Spend = 500, 300).
- It avoids distortion that mean would suffer in presence of extreme values.
- Suitable for skewed retail spend data.

A.iii Mean imputation (second technique, for comparison)

We can also compute means (alternative imputation):

Age mean (non-missing):

$$\text{Mean Age} = \frac{23 + 31 + 105 + 27 + 40 + 36 + 42}{7} = \frac{304}{7} \approx 43.43$$

Monthly_Spend mean (non-missing):

$$\text{Mean Spend} = \frac{20 + 35 + 500 + 25 + 45 + 38 + 300}{7} = \frac{963}{7} \approx 137.57$$

Purchase_Frequency mean (non-missing):

$$\text{Mean Frequency} = \frac{5 + 8 + 10 + 50 + 12 + 9 + 45}{7} = \frac{139}{7} \approx 19.86$$

If we used mean imputation:

- Age(C2) ≈ 43.43
- Monthly_Spend(C3) ≈ 137.57
- Purchase_Frequency(C5) ≈ 19.86

Why median is preferred over mean here:

- Mean is **pulled by outliers** (105, 500, 300), giving unrealistically high imputed values.
- Median gives **more realistic central values** for typical customers.
- For retail analytics, we want imputed values that reflect typical customer behaviour, not extremes.

B. Outlier detection and treatment

B.i Z-score method for Age and Monthly_Spend

We use:

$$z = \frac{x - \mu}{\sigma}$$

(Here we focus on detecting obvious extremes rather than exact thresholds.)

Age (non-missing): 23, 31, 105, 27, 40, 36, 42 Approximate mean and standard deviation:

- $\mu_{\text{Age}} \approx 43.43$
- $\sigma_{\text{Age}} \approx 25.92$

Z-score for Age = 105:

$$z_{105} = \frac{105 - 43.43}{25.92} \approx \frac{61.57}{25.92} \approx 2.38$$

Even though |z| < 3, **Age = 105** is **unrealistic for a retail customer**, so it is treated as an outlier using **domain knowledge**.

Monthly_Spend (non-missing): 20, 35, 500, 25, 45, 38, 300 Approximate mean and standard deviation:

- $\mu_{\text{Spend}} \approx 137.57$
- $\sigma_{\text{Spend}} \approx 174.50$

Z-score for Spend = 500:

$$z_{500} = \frac{500 - 137.57}{174.50} \approx \frac{362.43}{174.50} \approx 2.08$$

Z-score for Spend = 300:

$$z_{300} = \frac{300 - 137.57}{174.50} \approx \frac{162.43}{174.50} \approx 0.93$$

Again, **500 and 300** are **very high compared to typical spends (20–45)**, so they are treated as extreme values.

B.ii Identify records with extreme values

From the dataset:

- **C4:** Age = 105, Monthly_Spend = 500, Purchase_Frequency = 50
- **C8:** Monthly_Spend = 300, Purchase_Frequency = 45

These are **extreme** compared to other customers.

B.iii Treatment methods and justification

Common options:

1. **Capping (Winsorization)**
    - Set Age above a realistic maximum (e.g., 70) to 70.
    - Set Monthly_Spend above a threshold (e.g., 100) to 100.
2. **Transformation**
    - Apply log transformation to Monthly_Spend to reduce skewness.
3. **Removal**
    - Remove records that are clearly erroneous (e.g., Age = 105).

Example treatment:

- **Age:** Cap at 70
    - C4 Age: 105 → 70
- **Monthly_Spend:** Cap at 100
    - C4 Spend: 500 → 100
    - C8 Spend: 300 → 100

Justification:

- Age 105 is **not realistic** for typical retail customers → likely data entry error.
- Very high spends (500, 300) can **distort model training** and recommendation patterns.
- Capping preserves the records but **limits the influence of extremes**, which is useful when we don't want to lose data but want to control outlier impact.

In some strict settings, C4 could be **removed entirely** if considered invalid.

C. Categorical data standardization (Gender)

C.i Identify inconsistencies

Gender values in dataset:

Customer_ID	Gender
C1	M
C2	Male
C3	F
C4	Female
C5	male
C6	FEMALE
C7	M
C8	Female

Inconsistencies:

- Different cases: **Male, male, M**
- Different forms: **Female, FEMALE, F**

C.ii Standardize to "Male" / "Female"

We define mapping:

- **Male group:** M, Male, male → **Male**
- **Female group:** F, Female, FEMALE → **Female**

Standardized Gender:

Customer_ID	Original Gender	Standardized Gender
C1	M	Male
C2	Male	Male
C3	F	Female

Customer_ID	Original Gender	Standardized Gender
C4	Female	Female
C5	male	Male
C6	FEMALE	Female
C7	M	Male
C8	Female	Female

Justification:

- Machine learning models require **consistent categorical encoding**.
- Standardization avoids treating M and Male as different categories.
- It simplifies later steps like **one-hot encoding** or **label encoding**.

Q2: Answer:

A. Covariance Analysis

We compute covariance between **Income (₹K)** and **Purchase_Value (₹K)**.

Step 1: Extract the two variables

Cust_ID	Income (X)	Purchase_Value (Y)
U1	40	100
U2	60	150
U3	85	210
U4	120	320
U5	45	110
U6	75	180
U7	150	360
U8	70	160

n = 8

Step 2: Compute means

$$\bar{X} = \frac{40 + 60 + 85 + 120 + 45 + 75 + 150 + 70}{8} = \frac{645}{8} = 80.625$$

$$\bar{Y} = \frac{100 + 150 + 210 + 320 + 110 + 180 + 360 + 160}{8} = \frac{1590}{8} = 198.75$$

Step 3: Compute (Xi − X̄)(Yi − Ȳ)

Cust_ID	Xi	Yi	Xi − X̄	Yi − Ȳ	Product
U1	40	100	-40.625	-98.75	4016.41
U2	60	150	-20.625	-48.75	1005.47
U3	85	210	4.375	11.25	49.21
U4	120	320	39.375	121.25	4775.78
U5	45	110	-35.625	-88.75	3162.66
U6	75	180	-5.625	-18.75	105.47
U7	150	360	69.375	161.25	11178.28
U8	70	160	-10.625	-38.75	411.72

Sum of products:

$$\Sigma(Xi - \bar{X})(Yi - \bar{Y}) = 246,? \approx 246,? \text{ (sum of above)} = 246,?$$

Let's add them:

4016.41

- 1005.47 = 5021.88
- 49.21 = 5071.09
- 4775.78 = 9846.87
- 3162.66 = 13009.53
- 105.47 = 13115.00
- 11178.28 = 24293.28
- 411.72 = **24705.00**

Step 4: Covariance formula

$$\text{Cov}(X, Y) = \frac{\Sigma(Xi - \bar{X})(Yi - \bar{Y})}{n}$$

$$\text{Cov}(X, Y) = \frac{24705}{8} = 3088.125$$

A.ii Interpretation

- Covariance = **3088.125**
- Covariance > 0 → **positive relationship**

Meaning: As **Income increases**, **Purchase_Value also increases**. Higher-income customers tend to make higher-value purchases.

A.iii How covariance helps

Covariance helps identify:

- Whether two features move **together** (positive)
- Or move **opposite** (negative)
- Or are unrelated (near zero)

This helps detect **redundant features**, **multicollinearity**, and guides **feature selection**.

B. Feature Selection

B.i Identify redundant attributes

Using domain knowledge + correlation patterns:

- **Income** and **Purchase_Value** → strongly correlated
- **Orders** and **Avg_Balance** → both measure customer activity
- **EMI** may correlate with **Credit_Score** (financial stability)

Thus, redundancy exists.

B.ii Select optimal subset of features

Remove highly correlated or less useful attributes:

Recommended feature subset:

- Age
- Income
- Purchase_Value
- Credit_Score
- Orders
- Customer_Class (target)

Remove:

- Avg_Balance (redundant with Orders)
- EMI (redundant with Credit_Score)

B.iii Justification

- Reduces **multicollinearity**
- Improves **model speed** and **accuracy**
- Avoids overfitting
- Keeps only **high-impact predictors** for customer classification

C. Data Transformation

C.i Standardize dataset (except Cust_ID, Customer_Class)

We apply **z-score standardization**:

$$z = \frac{x - \mu}{\sigma}$$

Variables to standardize:

- Age
- Income
- Purchase_Value
- Credit_Score
- Orders
- Avg_Balance
- EMI

Example (Income):

Mean = 80.625 SD ≈ 38.9

For U1 (Income = 40):

$$z = \frac{40 - 80.625}{38.9} = -1.04$$

You would compute similarly for all values.

C.ii Why standardization is required

- Features like **Income (40–150)** and **Credit_Score (650–820)** have different scales
- ML models (KNN, SVM, Logistic Regression) perform poorly when scales differ
- Standardization ensures:
    - Equal weight for all features
    - Faster convergence
    - Better model stability
    - Improved accuracy

Q3: Answer

A. Equal-Width Binning

We apply **3 bins** for:

- Study_Hours
- Test_Score

A.i Study_Hours – Equal-Width Binning

Step 1: Identify min and max

Study_Hours values: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

- Min = 2
- Max = 11

Step 2: Compute bin width

$$\text{Bin Width} = \frac{\text{Max} - \text{Min}}{3} = \frac{11 - 2}{3} = 3$$

Step 3: Define intervals

- **Bin 1:** 2 – 5
- **Bin 2:** 5 – 8
- **Bin 3:** 8 – 11

Step 4: Assign Study_Hours to bins

Student_ID	Study_Hours	Bin
S1	2	Bin 1
S2	3	Bin 1
S3	4	Bin 1
S4	5	Bin 2
S5	6	Bin 2
S6	7	Bin 2
S7	8	Bin 3
S8	9	Bin 3

Student_ID	Study_Hours	Bin
S9	10	Bin 3
S10	11	Bin 3

A.ii Test_Score – Equal-Width Binning

Test_Score values: 40, 50, 55, 65, 75, 85, 90, 95, 98, 100

- Min = 40
- Max = 100

Step 1: Compute bin width

$$\text{Bin Width} = \frac{100 - 40}{3} = 20$$

Step 2: Define intervals

- **Bin 1:** 40 – 60
- **Bin 2:** 60 – 80
- **Bin 3:** 80 – 100

Step 3: Assign Test_Score to bins

Student_ID	Test_Score	Bin
S1	40	Bin 1
S2	50	Bin 1
S3	55	Bin 1
S4	65	Bin 2
S5	75	Bin 2
S6	85	Bin 3
S7	90	Bin 3
S8	95	Bin 3
S9	98	Bin 3
S10	100	Bin 3

B. Equal-Frequency Binning

We divide data into **3 bins with equal number of records**.

Total records = 10 Records per bin = 10 ÷ 3 ≈ 3 or 4

We will use:

- Bin 1 → 3 records
- Bin 2 → 3 records
- Bin 3 → 4 records

B.i Study_Hours – Equal-Frequency

Sorted Study_Hours: 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Bins

- **Bin 1:** 2, 3, 4
- **Bin 2:** 5, 6, 7
- **Bin 3:** 8, 9, 10, 11

B.ii Test_Score – Equal-Frequency

Sorted Test_Score: 40, 50, 55, 65, 75, 85, 90, 95, 98, 100

Bins

- **Bin 1:** 40, 50, 55
- **Bin 2:** 65, 75, 85
- **Bin 3:** 90, 95, 98, 100

C. Concept Hierarchy Construction

Concept hierarchies convert continuous values into **interpretable categories**.

C.i Study_Hours → Low / Medium / High

We define:

- **Low:** 2–4 hours
- **Medium:** 5–7 hours
- **High:** 8–11 hours

Student_ID	Study_Hours	Category
S1	2	Low
S2	3	Low

Student_ID	Study_Hours	Category
S3	4	Low
S4	5	Medium
S5	6	Medium
S6	7	Medium
S7	8	High
S8	9	High
S9	10	High
S10	11	High

C.ii Test_Score → Poor / Average / Excellent

We define:

- **Poor:** 40–60
- **Average:** 60–80
- **Excellent:** 80–100

Student_ID	Test_Score	Category
S1	40	Poor
S2	50	Poor
S3	55	Poor
S4	65	Average
S5	75	Average
S6	85	Excellent
S7	90	Excellent
S8	95	Excellent
S9	98	Excellent
S10	100	Excellent

Q4 Answer:

Given data: **200, 300, 400, 600, 1000**

We apply:

1. **Min–Max Normalization (range 0 to 1)**
2. **Z-Score Normalization**

(a) Min–Max Normalization (0 to 1)

Formula:

$$x' = \frac{x - \min}{\max - \min}$$

Given:

- min = 200
- max = 1000
- range = 1000 − 200 = 800

Now normalize each value:

1. For 200

$$\frac{200 - 200}{800} = 0$$

2. For 300

$$\frac{300 - 200}{800} = \frac{100}{800} = 0.125$$

3. For 400

$$\frac{400 - 200}{800} = \frac{200}{800} = 0.25$$

4. For 600

$$\frac{600 - 200}{800} = \frac{400}{800} = 0.5$$

5. For 1000

$$\frac{1000 - 200}{800} = 1$$

Min–Max Normalized Values

Original	**Normalized (0–1)**
200	0
300	0.125

Original	Normalized (0–1)
400	0.25
600	0.5
1000	1

Q5: Answer:

A. Equal-Frequency Binning

We divide the dataset into **3 bins** with approximately equal number of records.

Total records = 10 Records per bin ≈ 10 ÷ 3 = 3 or 4

A.i Sort Visit_Duration

Sorted Visit_Duration (seconds):

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

A.ii Create 3 equal-frequency bins

- **Bin 1:** first 3 values → 15, 30, 45
- **Bin 2:** next 3 values → 60, 120, 300
- **Bin 3:** last 4 values → 600, 900, 1800, 3600

A.iii Assign each value to its bin

Customer_ID	Visit_Duration	Bin
C1	15	Bin 1
C2	30	Bin 1
C3	45	Bin 1
C4	60	Bin 2
C5	120	Bin 2
C6	300	Bin 2
C7	600	Bin 3

Customer_ID	Visit_Duration	Bin
C8	900	Bin 3
C9	1800	Bin 3
C10	3600	Bin 3

A.iv Effectiveness Analysis

- **Bin 1** groups *very short visits* (15–45 sec).
- **Bin 2** groups *moderate visits* (1–5 minutes).
- **Bin 3** groups *long visits* (10 minutes to 1 hour).

Equal-frequency binning **clearly separates short, medium, and long browsing behavior**, making segmentation more meaningful.

B. Z-Score Normalization

Formula:

$$z = \frac{X - \mu}{\sigma}$$

B.i Compute Mean (μ)

$$\mu = \frac{15 + 30 + 45 + 60 + 120 + 300 + 600 + 900 + 1800 + 3600}{10}$$

Sum = 7470

$$\mu = 7470/10 = 747$$

B.ii Compute Standard Deviation (σ)

We compute squared deviations:

X	X − μ	(X − μ)²
15	-732	535,824
30	-717	514,089
45	-702	492,804
60	-687	471,969
120	-627	393,129

X	X − μ	(X − μ)²
300	-447	199,809
600	-147	21,609
900	153	23,409
1800	1053	1,108,809
3600	2853	8,142,609

Sum of squares = 11,903,060

Variance:

$$\sigma^2 = 11{,}903{,}060/10 = 1{,}190{,}306$$

Standard deviation:

$$\sigma = \sqrt{1{,}190{,}306} \approx 1090.12$$

B.iii Apply Z-Score Normalization

$$z = \frac{X - 747}{1090.12}$$

X	Z-Score
15	-0.67
30	-0.66
45	-0.64
60	-0.63
120	-0.57
300	-0.41
600	-0.14
900	0.14
1800	0.97
3600	2.62

B.iv Handling Extreme Values

- Extreme durations (1800, 3600 sec) produce **large positive z-scores**.
- Z-score normalization **does not compress outliers**; it highlights them.

- This is useful for clustering because extreme users stand out clearly.

C. Min-Max Normalization

Formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Min = 15 Max = 3600 Range = 3585

C.i Apply Min-Max Normalization

X	**X' = (X − 15)/3585**
15	0
30	0.004
45	0.008
60	0.013
120	0.029
300	0.079
600	0.163
900	0.247
1800	0.498
3600	1

C.ii Comparison with Z-Score

Method	**Behavior**	**Sensitivity to Outliers**
Z-Score	Centers data around mean	**High** — extreme values get large z-scores
Min-Max	Scales data to [0,1]	**Very sensitive** — outliers stretch the scale
Equal-Frequency	Groups values evenly	**Low sensitivity** — bins remain balanced

Conclusion:

- Min-Max is best for algorithms needing fixed range (NN, K-means).
- Z-score is best when data has extreme values and needs centering.
- Equal-frequency is best for segmentation and interpretability.